

Jingtian Wu

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Education

Cornell University

Ithaca, NY

B.A. in Computer Science — GPA: 3.97/4.00

Fall 2022–Spring 2026

Research Interests: Multimodality, Natural Language Processing, Reinforcement Learning, Machine Learning

Relevant Coursework: NLP, RL, ML, Computer Vision, Algorithm, AI Foundations, AI Philosophy, OOP & Data Structures, Functional Programming, Linear Algebra, Discrete Math, Calculus, Computer Systems

Research

Behavioral Evaluation and Control in AI-Simulated Patients

Researcher advised by Yann Hicke and Prof. Claire Cardie

Summer 2025 Ongoing

- Investigating behavioral consistency in LLMs used as AI-simulated patients (AI-SPs) for medical education.
- Developing methods to detect and control these behaviors using supervised classifiers and reinforcement learning, enabling more realistic and pedagogically aligned simulations.

OmniWizz — Multimodal Inspirational Tool

Researcher and Lead Developer

Summer 2025 Ongoing

- Designed a unified system for cross-modal creativity, allowing users to input any combination of text, image, or audio and receive inspiring outputs across modalities. [Link to OmniWizz]

Automated Fine-Tuning Dataset Creation via Knowledge Map Expansion

Independent Researcher advised by Claire Cardie

Spring 2025

- Developed an automated LLM-based framework that constructs hierarchical medical knowledge maps and generates multiple-choice QA datasets for each leaf node to support efficient self-supervised fine-tuning.

Culturally and Temporally Contextualized Lyric Generation

Researcher advised by Matthew Wilkens

Spring 2025

- Developing a lyric generation system that adapts to cultural and temporal contexts.

Automatic Code Generation System

Independent Researcher advised by Prof. Claire Cardie and Wenting Zhao

Fall 2024

- Developed an automated code generation system using Aider on the “Commit-0” benchmark and explored multi-agent collaboration, rollback mechanisms, curriculum learning, and tree search-based planning to improve its performance.

Reasoning Court (RC) Framework for Enhancing Accuracy in Multi-Hop Question Datasets

BURE Researcher advised by Prof. Claire Cardie and Wenting Zhao

Summer 2024

- Developed a framework that improves LLM performance on multi-hop reasoning and fact-verification benchmarks, achieving an average absolute improvement of 5 percentage points over previous best-performing few-shot prompting baselines.

Computer Music Synthesis and Composition

Researcher advised by Prof. Roger B. Dannenberg

Summer 2023

- Researched algorithmic composition, digital audio theory, sound synthesis algorithms, and Nyquist computations.

Projects

REVIEWER2: A Two-Stage LLM Framework for Academic Peer Review Generation

Collaboration with Prof. Thorsten Joachims and PhD student Zhaolin Gao

- Main lead in developing a demo system for REVIEWER2, a two-stage LLM framework for academic peer review generation.

Activities

BOOM (BITS ON OUR MINDS) 2024

- Presented “REVIEWER2” project to Cornell community and industry and featured in the **Cornell Chronicle**.

Teaching Experience

Fall 2024: TA for CS 3700 — **Foundations of AI Reasoning and Decision-Making**, Cornell CIS

Fall 2025: TA for CS 3700 — **Foundations of AI Reasoning and Decision-Making**, Cornell CIS

Publications

2025: Jingtian Wu, Claire Cardie, “Reasoning Court: Combining Reasoning, Action, and Judgment for Multi-Hop Reasoning,” arXiv:2504.09781

2023: Jingtian Wu, "Algorithmic Composition of Music Utilizing the Digits of Pi," Computer Science and Intelligent Communication (CSIC 2023)

Skills

Programming: Python, JavaScript, Java, OCaml, $\text{\LaTeX}$ **Web Development:** HTML/CSS, Node.js, Flask, FastAPI, React **Frameworks/Libraries:** PyTorch, React **Tools:** Visual Studio Code, Jupyter, GitHub